

AKSHAY KUMAR VERMA

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SUMMARY

Highly motivated Data Science professional with proven expertise in machine learning, deep learning, and data-driven problem-solving. M. Tech graduate from IIT Roorkee and B. Tech graduate from IIT Palakkad, with hands-on experience in developing scalable ML models, optimizing algorithms, and deploying applications for real-world challenges. Demonstrated ability to work both independently and collaboratively on diverse projects, such as EV route optimization, power theft mitigation, and NLP-based drug recommendation systems. Strong communicator with a track record of delivering impactful solutions in research and industry settings.

WORK EXPERIENCE

Research Assistant, IIT Roorkee

Jul 2024 - Dec 2024

- Developed Python-based algorithms to optimise bus routes for EV deployment, reducing costs and improving efficiency.
- Conducted in-depth data analysis, achieving a diesel-to-electric bus replacement ratio of 1.03.
- Developed an algorithm to identify optimal routes for electric buses.
- Designed optimisation models to improve fleet utilisation and reduce emissions.
- Tools/Technologies:** Python, Linear Programming, Data Visualisation (Power BI), Pandas, Excel.

Jun 2023 - Jul 2023

Intern, Magnus IT Solution pvt ltd

- Analysed comprehensive power consumption data to identify theft patterns in Uttar Pradesh.
- Built dashboards using Power BI for real-time decision-making and trend analysis.
- Tools/Technologies:** Python, Power BI, SQL, Excel.

PROJECTS

Drug Recommendation system using NLP

- Designed a Flask-based application to recommend drugs based on patient symptoms using the facebook/bart-large-mnli model for natural language inference and a modular knowledge base for symptom-to-condition mapping.
- Optimized system performance by implementing entailment thresholds ($\geq 75\%$) and enhancing server efficiency to handle the large model size.
- Delivered a user-friendly interface for structured drug recommendations, enabling efficient preliminary diagnosis and treatment planning.
- Tools/Technologies:** Python, Flask, Facebook BART, Pandas.

Electric Vehicle remaining range prediction

- Developed several machine learning and deep learning models to predict the power consumption of EVs.
- Developed a web application to deploy the ML model to predict the remaining range of EVs.
- Tools/Technologies:** TensorFlow, Flask, Python, Pandas, Scikit-learn.

Mathematical model on optimization of parking capacity

- Designed and implemented a mathematical model based on linear programming techniques to optimize the parking capacity of a designated area.
- Tools/Technologies:** Python, NumPy, Linear Programming.

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, Flask
- Data Science and ML:** Predictive Analytics, Statistical Modeling, Deep Learning, NLP
- Frameworks/Libraries:** TensorFlow, Scikit-learn, Pandas, NLTK
- Data Visualization:** Power BI, Matplotlib, Seaborn
- Tools:** MySQL, Git, Jupyter Notebook, Excel
- Soft Skills:** Communication, Collaboration, Problem-Solving

EDUCATION

M. Tech in Infrastructure Systems

Indian Institute of Technology, Roorkee
8.526 CGPA

2022 - 2024

B. Tech in Civil Engineering

Indian Institute of Technology, Palakkad
7.56 CGPA

2015 - 2019

Senior secondary school

Lucknow Public School
92.4 %

2014 - 2015

CERTIFICATIONS

- Machine Learning with Python - IBM (Coursera)
- Python for Data Science and Machine Learning Masterclass - Udemy (Online Course)
- Complete TensorFlow 2 and Keras Deep Learning Bootcamp - Udemy (Online Course)
- NLP - Natural Language Processing with Python - Udemy (Online Course)